

Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.70_e-1_d1)

Metric

$$ds^2 = \frac{e^{(2\gamma)}}{f} du \otimes du - \frac{e^{(2\gamma)}}{f} dr \otimes dr + f dv \otimes dv + uw f dv \otimes dy + uw f dy \otimes dv + \left(\frac{u^2 w^2 f^2 + r^2}{f} \right) dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, v, y)$$

Metric Functions

$$f(r) = f$$

$$\gamma(r) = \gamma$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}e^\gamma}{2\sqrt{f}} \right) du + \left(\frac{\sqrt{2}e^\gamma}{2\sqrt{f}} \right) dr$$

$$n = \left(-\frac{\sqrt{2}e^\gamma}{2\sqrt{f}} \right) du + \left(\frac{\sqrt{2}e^\gamma}{2\sqrt{f}} \right) dr$$

$$m = \left(\sqrt{\frac{1}{2}}\sqrt{f} \right) dv + \left(\sqrt{\frac{1}{2}}uw\sqrt{f} + \frac{i\sqrt{2}r}{2\sqrt{f}} \right) dy$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}} \sqrt{f} \right) dv + \left(\sqrt{\frac{1}{2}} uw \sqrt{f} - \frac{i \sqrt{2} r}{2 \sqrt{f}} \right) dy$$

Contravariant components

$$l = \left(\frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \right) du + \left(-\frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \right) du + \left(-\frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \right) dr$$

$$m = \left(-\frac{i \sqrt{2} uw \sqrt{f}}{2 r} + \frac{\sqrt{2}}{2 \sqrt{f}} \right) dv + \left(\frac{i \sqrt{2} \sqrt{f}}{2 r} \right) dy$$

$$\overline{m} = \left(\frac{i \sqrt{2} uw \sqrt{f}}{2 r} + \frac{\sqrt{2}}{2 \sqrt{f}} \right) dv + \left(-\frac{i \sqrt{2} \sqrt{f}}{2 r} \right) dy$$

Directional Derivatives

$$D = \frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \partial_u - \frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \partial_u - \frac{1}{2} \sqrt{2} e^{(-\gamma)} \sqrt{f} \partial_r$$

$$\delta = -\frac{i \sqrt{2} uw \sqrt{f}}{2 r} + \frac{\sqrt{2}}{2 \sqrt{f}} \partial_v + \frac{i \sqrt{2} \sqrt{f}}{2 r} \partial_y$$

$$\bar{\delta} = \frac{i \sqrt{2} uw \sqrt{f}}{2 r} + \frac{\sqrt{2}}{2 \sqrt{f}} \partial_v - \frac{i \sqrt{2} \sqrt{f}}{2 r} \partial_y$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}e^{(-\gamma)}\sqrt{f}}{4r}$$

$$\sigma = -\frac{i\sqrt{2}we^{(-\gamma)}f^{\frac{3}{2}}}{4r} + \frac{\sqrt{2}e^{(-\gamma)}\frac{\partial}{\partial r}f}{4\sqrt{f}} - \frac{\sqrt{2}e^{(-\gamma)}\sqrt{f}}{4r}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}e^{(-\gamma)}\sqrt{f}}{4r}$$

$$\lambda = \frac{i\sqrt{2}we^{(-\gamma)}f^{\frac{3}{2}}}{4r} - \frac{\sqrt{2}e^{(-\gamma)}\frac{\partial}{\partial r}f}{4\sqrt{f}} + \frac{\sqrt{2}e^{(-\gamma)}\sqrt{f}}{4r}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{i\sqrt{2}we^{(-\gamma)}f^{\frac{3}{2}}}{8r} - \frac{1}{4}\sqrt{2}e^{(-\gamma)}\sqrt{f}\frac{\partial}{\partial r}\gamma + \frac{\sqrt{2}e^{(-\gamma)}\frac{\partial}{\partial r}f}{8\sqrt{f}}$$

$$\gamma = \frac{i\sqrt{2}we^{(-\gamma)}f^{\frac{3}{2}}}{8r} + \frac{1}{4}\sqrt{2}e^{(-\gamma)}\sqrt{f}\frac{\partial}{\partial r}\gamma - \frac{\sqrt{2}e^{(-\gamma)}\frac{\partial}{\partial r}f}{8\sqrt{f}}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{w^2 e^{(-2\gamma)} f^3}{8 r^2} - \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f^2}{8 f} + \frac{e^{(-2\gamma)} f \frac{\partial}{\partial r} \gamma}{2 r}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{w^2 e^{(-2\gamma)} f^3}{4 r^2} - \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f^2}{4 f} + \frac{1}{4} e^{(-2\gamma)} \frac{\partial^2}{(\partial r)^2} f + \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f}{4 r}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{w^2 e^{(-2\gamma)} f^3}{16 r^2} - \frac{1}{4} e^{(-2\gamma)} f \frac{\partial^2}{(\partial r)^2} \gamma - \frac{3 e^{(-2\gamma)} \frac{\partial}{\partial r} f^2}{16 f} + \frac{1}{8} e^{(-2\gamma)} \frac{\partial^2}{(\partial r)^2} f + \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f}{8 r}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{w^2 e^{(-2\gamma)} f^3}{4 r^2} - \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f^2}{4 f} + \frac{1}{4} e^{(-2\gamma)} \frac{\partial^2}{(\partial r)^2} f + \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f}{4 r}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{w^2 e^{(-2\gamma)} f^3}{8 r^2} - \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f^2}{8 f} + \frac{e^{(-2\gamma)} f \frac{\partial}{\partial r} \gamma}{2 r}$$

$$\Lambda = -\frac{w^2 e^{(-2\gamma)} f^3}{48 r^2} + \frac{1}{12} e^{(-2\gamma)} f \frac{\partial^2}{(\partial r)^2} \gamma + \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f^2}{16 f} - \frac{1}{24} e^{(-2\gamma)} \frac{\partial^2}{(\partial r)^2} f - \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f}{24 r}$$

Weyl Tensor Components

$$\Psi_0 = \frac{w^2 e^{(-2\gamma)} f^3}{4r^2} - \frac{i w e^{(-2\gamma)} f^2 \frac{\partial}{\partial r} \gamma}{2r} + \frac{3i w e^{(-2\gamma)} f \frac{\partial}{\partial r} f}{4r} + \frac{1}{2} e^{(-2\gamma)} \frac{\partial}{\partial r} f \frac{\partial}{\partial r} \gamma - \frac{i w e^{(-2\gamma)} f^2}{4r^2} - \frac{1}{4} e^{(-2\gamma)} \frac{\partial^2}{(\partial r)^2} f - \frac{e^{(-2\gamma)} f \frac{\partial}{\partial r} \gamma}{2r}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{w^2 e^{(-2\gamma)} f^3}{12r^2} - \frac{i w e^{(-2\gamma)} f \frac{\partial}{\partial r} f}{4r} - \frac{1}{6} e^{(-2\gamma)} f \frac{\partial^2}{(\partial r)^2} \gamma + \frac{i w e^{(-2\gamma)} f^2}{4r^2} + \frac{1}{12} e^{(-2\gamma)} \frac{\partial^2}{(\partial r)^2} f - \frac{e^{(-2\gamma)} \frac{\partial}{\partial r} f}{6r}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{w^2 e^{(-2\gamma)} f^3}{4r^2} - \frac{i w e^{(-2\gamma)} f^2 \frac{\partial}{\partial r} \gamma}{2r} + \frac{3i w e^{(-2\gamma)} f \frac{\partial}{\partial r} f}{4r} + \frac{1}{2} e^{(-2\gamma)} \frac{\partial}{\partial r} f \frac{\partial}{\partial r} \gamma - \frac{i w e^{(-2\gamma)} f^2}{4r^2} - \frac{1}{4} e^{(-2\gamma)} \frac{\partial^2}{(\partial r)^2} f - \frac{e^{(-2\gamma)} f \frac{\partial}{\partial r} \gamma}{2r}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"